

YINCHUAN, China

WMO: 536140

Lat: 38.4714N

Lon: 106.2078E

Elev: 3648

StdP: 12.86

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 3.5	(c) 7.0	(d) -14.2	(e) 2.9	(f) 17.8	(g) -10.2	(h) 3.6	(i) 18.6	(j) 20.4	(k) 32.3	(l) 15.5	(m) 28.4	(n) 3.7	(o) 0	(p) 0.444

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 18.6	(c) 91.3	(d) 65.6	(e) 89.0	(f) 65.3	(g) 86.7	(h) 64.5	(i) 71.7	(j) 83.2	(k) 70.1	(l) 81.8	(m) 68.7	(n) 80.3	(o) 5.8	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.3	119.1	77.6	66.5	111.8	76.8	64.8	105.2	75.5	38.2	83.0	36.7	82.0	35.3	80.2	79.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
17.3	13.5	11.0	-1.0	96.2	4.2	2.3	-4.0	97.8	-6.5	99.1	-8.8	100.4	-11.9	102.0	
			DB												
			WB	-2.2	74.0	3.9	2.6	-5.0	75.9	-7.2	77.4	-9.4	78.9	-12.2	80.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.1	20.7	28.9	42.1	55.8	64.8	72.7	76.3	73.2	64.0	51.7	37.1	24.1		
	DBStd	20.25	6.54	8.11	8.71	7.92	6.16	4.78	4.39	4.79	5.69	6.91	7.43	6.52		
	HDD50	3059	908	591	271	35	1	0	0	0	0	63	388	802		
	HDD65	6072	1373	1011	711	290	80	5	0	3	83	413	836	1267		
	CDD50	3444	0	0	26	207	459	681	814	718	421	116	2	0		
	CDD65	982	0	0	0	12	74	236	349	256	54	1	0	0		
	CDH74	8956	0	0	6	181	811	2211	3265	2056	403	23	0	0		
	CDH80	3108	0	0	0	41	205	789	1317	675	80	1	0	0		
(14) Wind	WSAvg	4.9	4.3	4.8	5.8	6.1	5.8	5.4	4.8	4.6	4.3	4.3	4.6	4.3		
(15) Precipitation	PrecAvg	7.8	0.0	0.1	0.2	0.4	0.8	0.8	1.7	1.9	1.0	0.5	0.2	0.0		
	PrecMax	14.0	0.3	0.5	1.1	1.9	3.0	3.5	5.9	5.1	3.9	2.0	1.0	0.3		
	PrecMin	3.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.2	0.0	0.0	0.0	0.0		
	PrecStd	2.6	0.1	0.1	0.3	0.5	0.7	0.7	1.3	1.1	0.8	0.4	0.3	0.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.3	56.3	72.6	84.8	88.6	93.1	95.4	92.5	87.3	77.1	62.7	47.6		
		MCWB	32.0	38.8	47.8	55.1	57.9	61.8	67.2	67.5	62.9	54.7	44.4	33.8		
	2%	DB	39.5	50.2	66.0	79.0	85.1	90.1	91.9	89.1	81.8	72.3	58.2	41.6		
		MCWB	29.2	35.9	44.9	51.9	57.0	62.3	67.3	67.3	60.5	53.4	42.6	30.9		
	5%	DB	35.8	46.3	61.6	74.6	82.0	87.4	89.6	86.6	78.4	68.4	53.9	38.2		
		MCWB	26.7	34.0	42.5	50.2	56.6	61.8	67.3	66.6	59.5	51.3	40.8	29.3		
	10%	DB	32.3	42.7	57.1	70.6	78.7	84.8	87.2	83.9	75.4	64.6	49.6	35.0		
		MCWB	24.6	31.7	40.5	48.6	55.2	61.6	66.7	65.9	59.5	49.7	38.9	27.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.4	40.3	49.4	58.1	63.0	69.3	74.4	73.7	68.7	58.7	47.1	34.8		
		MCDB	44.3	52.4	67.4	73.5	79.1	82.5	87.1	84.8	75.4	71.1	57.2	44.9		
	2%	WB	29.3	36.9	46.1	55.1	60.9	67.3	72.3	71.4	65.5	56.1	44.0	31.4		
		MCDB	38.9	48.5	62.7	71.6	76.8	80.8	85.2	81.6	74.7	67.2	54.8	39.9		
	5%	WB	27.1	34.5	43.6	52.8	59.0	65.8	70.7	70.0	63.3	53.8	42.0	29.7		
		MCDB	35.1	45.2	59.2	68.6	74.6	79.3	83.3	80.7	72.8	64.0	51.6	37.4		
	10%	WB	24.9	32.2	41.3	50.6	57.4	64.3	69.3	68.6	61.4	51.6	39.8	27.8		
		MCDB	32.0	41.8	55.0	67.5	73.1	78.3	81.7	79.5	71.0	61.2	48.4	34.6		
(35) Mean Daily Temperature Range	MDBR		19.0	21.4	21.4	22.3	21.7	20.2	18.6	18.0	18.8	21.1	17.7	16.9		
	5% DB	MCDBR	23.5	25.7	27.6	28.7	27.5	25.2	22.9	22.2	24.3	25.8	24.2	20.9		
		MCWBR	15.4	15.1	13.3	11.2	10.0	8.1	7.2	7.3	8.6	10.9	12.4	13.1		
	5% WB	MCDBR	22.7	24.6	25.3	23.1	21.7	20.0	18.3	16.8	17.9	22.0	21.6	19.4		
		MCWBR	15.1	14.7	12.8	10.1	9.1	7.9	7.8	7.2	7.9	10.3	11.7	12.4		
(40) Clear-Sky Solar Irradiance	taub		0.353	0.410	0.464	0.504	0.522	0.529	0.510	0.513	0.492	0.446	0.382	0.345		
	taud		2.081	1.946	1.847	1.794	1.819	1.866	1.991	1.959	1.940	1.969	2.061	2.115		
	Ebn at Noon		250	249	248	247	246	244	247	243	238	234	236	243		
	Edh at Noon		37	49	60	67	67	64	56	56	54	47	38	34		
(44) All-Sky Solar Radiation	RadAvg		896	1192	1587	1902	2113	2085	1965	1774	1464	1289	984	824		
	RadStd		82	75	91	115	105	144	98	97	120	81	80	61		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.92	N/A	-2.21	+1.10	-0.92	-1.84	N/A	-268	+220	+113
(47) Regional (4 neighbors)		N/A	N/A	N/A	N/A	N/A	-1.04	N/A	N/A	N/A	N/A

Nomenclature: See separate page